# What pollen can tell us about past landscapes

## Bijux Pollenomics: results, limits and decisions for collaborators

Bijux Pollenomics

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Source-bound research briefing

# The question: what grew around a dated sediment core?

- Sediment preserves grains from plants near and far. Their counts record pollen; we want to understand vegetation and landscape change.
- A useful result must identify the physical core, sample age, counted taxa and the landscape area represented. Those decisions determine whether past forest or farming change can be interpreted.
- The project traces those inputs, checks selected published calculations and tests predictions on declared pollen targets.

We currently have observed-pollen prediction and bounded calculation evidence. We have no independent measured-vegetation accuracy score for our Nordic predictions.

Evidence: `reports/science/source-cohorts.md`; `reports/science/vegetation-reference.md`

# Grain share and land-cover share are different quantities

| What we observe | What a model tries to infer |
|---|---|
| Pollen fraction: a taxon's grains divided by a declared grain count. | Vegetation fraction: a taxon's cover divided by a declared landscape area. |

Plants produce unequal quantities of pollen; grains travel and settle differently. A lake's size and type affect the incoming signal.

Productivity and transport corrections connect these quantities under assumptions. A grain percentage alone is not a measured vegetation percentage.

Evidence: `reports/science/landclim-paper.md`; `reports/science/loveoptim-paper.md`

# Regional and local reconstruction need different inputs

- **REVEALS**: Regional Estimates of VEgetation Abundance from Large Sites. Large-site pollen informs regional cover under declared transport and basin assumptions.
- **LOVE**: LOcal Vegetation Estimates. Regional background and small-site pollen inform cover in a defined local area.
- Together they form the **Landscape Reconstruction Algorithm (LRA)**. Relative pollen productivity estimates (PPEs), fall speed, basin geometry and selected ages are inputs, not optional annotations.

A working implementation does not admit a new historical core. Its source identity, parameters, geometry, rights and physical interpretation need evidence.

Evidence: `reports/science/reveals-theory-paper.md`; `reports/science/love-theory-paper.md`

# A concrete source-to-result example: one lake sample

1. White Horse Hill (`WHH05`), depth 16: source age 1,320 calibrated years before 1950, with 301 grains in named source fields.
2. Four explicitly excluded grains leave 297 on the author's declared axis. The age enters [1,250, 1,750), labelled 1,500 years BP.
3. On this prepared example, author R and our implementation give modeled *Calluna* cover 0.924883 and standard error 0.046807, in fraction units.

The interval label is not a new date. This is a modeled site value, not 92.49% of the grains or measured vegetation. Numerical agreement checks this calculation.

Evidence: `reports/science/landclim-paper.md`; `reports/science/publication-correspondence/author-example-site-replay.md`

# Three checks answer three different questions

| Check | Compare | Conclusion it can support |
|---|---|---|
| Calculation | Our numbers and source R on identical inputs. | Selected numerical methods correspond. |
| Pollen prediction | Frozen predictions and withheld observed pollen. | Pollen error on the tested cohort. |
| Vegetation accuracy | Inferred cover and independent measured cover with matched support. | Ecological accuracy on admitted pairs. |

The first two exist within bounded scopes. The third remains unavailable for our Nordic predictions; another model's cover is not an independent measurement.

Evidence: `reports/science/method-correspondence.md`; `reports/science/pollen-prediction.md`; `reports/science/vegetation-reference.md`

# Nordic pollen predictions: coverage and error differ

| Target | Sites | Samples | Bins | Lowest MAE; method |
|---|---|---|---|---|
| Denmark | 1 | 91 | 24/24 | 0.023242; nearest |
| Finland | 28 | 1,797 | 435/435 | 0.017335; nearest |
| Norway | 52 | 3,544 | 1,084/1,178 | 0.019459; IDW |
| Sweden | 3 | 228 | 51/54 | 0.028740; IDW |

Bins show scored/attempted support. IDW weights training sites by inverse squared distance; nearest uses the nearest eligible site. The target country is excluded from training. MAE is mean absolute error in pollen fractions on 31 author targets, aggregated equally across sites; lower is better within this declared comparison. A value 0.02 is two percentage points of mean fraction error.

Previously viewed development outcomes; method order is descriptive. Denmark has one site. Norway has 94 unsupported bins and Sweden three; these do not become zero-error successes.

Evidence: `reports/science/denmark-finland-norway-author-pollen.md`; `reports/science/sweden-author-pollen.md`

# Age support can change which observations are scored

- At Sämbosjön, the 16-label observed-pollen study scored 22 bins from 111 samples. Source age intervals cross bin boundaries for 58 samples.
- Keeping only contained intervals leaves 53 samples and 21 bins. On identical predictions, primary MAE is 0.023054 versus 0.022881 for original counts on those same bins.
- One bin has no contained sample and retains refused, null scores. Retained samples contribute 185,150 native grains, including 108,163 on the 16-label response.

One exposed core; the source bounds have no stated confidence level or joint age posterior. This sensitivity does not establish independent confirmation or vegetation accuracy.

Evidence: `reports/science/sambosjon-direct-pollen.md`; `docs/report/regions/nordic/sambosjon-direct-pollen/interval-support/report.json`

# English LRA demonstrates a bounded executable workflow

- Start from 15 author sites in the 1,250–1,750 BP interval, with 30 pollen targets after a declared transport exclusion.
- Hold out one site at a time, reconstruct with the remaining sites, and freeze three forward-pollen predictions for that target.
- Fourteen sites produce all three predictions. One fold refuses after three training optimizations fail.

This checks reconstruction-to-pollen prediction on English author examples. No independent measured cover is available, and this does not admit Nordic LRA inputs.

Evidence: `reports/science/english-lra-pollen-prediction.md`

# On the same English targets, direct pollen error is lower

| | Pollen MAE | | Hellinger distance | |
|---|---|---|---|---|
| Method | Direct | LRA forward | Direct | LRA forward |
| Training mean | .020496 | .026273 | .282082 | .360671 |
| Nearest site | .019552 | .021132 | .255767 | .286001 |
| Inverse distance squared | .018579 | .021088 | .243673 | .285415 |

Fourteen matched sites, 30 taxa; both metrics average equally across sites. Hellinger compares compositions on a 0–1 scale; zero means identical, and lower is better.

Training information differs between routes and the outcomes are exposed development evidence. Lower pollen error does not establish a better vegetation reconstruction.

Evidence: `reports/science/pollen-prediction.md`;

`docs/report/pollen-reconstruction/author-lra-constrained/direct-pollen/paired-hellinger.json`

# Matching author code does not reproduce the whole archive

- Selected LandClim examples: author R and our implementation agree on 14,074 supported means and standard errors. The largest fraction differences are $1.84 \times 10^{-13}$ and $7.16 \times 10^{-14}$.
- Only one of 25 time windows has identical boundaries in example and archive: 100–350 BP. Archived mean cover differs by up to 9.20 and 10.55 percentage points in the two grids.
- The released example has 17 sites; the paper selected 1,128 historical records. Their original input revisions and archive functional-type transformations remain unverified here.

The archive gap is retained, but its cause is not identified. Numerical parity, historical source identity and ecological accuracy are separate questions.

Evidence: `reports/science/landclim-paper.md`; `reports/science/publication-correspondence/published-window-provenance.md`

# Published vegetation references have useful limits

- Historical Småland maps provide coverage and pollen anchors; the numerical class-area fractions needed for a new score are unavailable.
- A southern Swedish published figure supports nine rounded label comparisons. It does not recover all original taxa or score our model.
- Namibian observations join ten soil-pollen sites to 15 vegetation-radius labels. Repeated radius rows are not new pollen samples, and the population differs from Nordic lakes.

Our Nordic plot/pollen screen admits zero of 16 nearby leads. None of these distinct sources supplies independent measured cover for our current Nordic predictions.

Evidence: `reports/science/vegetation-reference.md`; `reports/science/paper-accounts.md`

# What collaborators can inspect and decide

1. Read the canonical paper account: question, concepts, worked example, original findings and our separate result.
2. Follow its exact table and source identities; use the reproduction route with lawful source access.
3. Help resolve a named core/age/basin/count/parameter question, historical archive transformation, or matched vegetation reference.
4. Review whether assumptions and spatial/time support justify the intended scientific conclusion. Actual reader feedback and qualified decisions remain open.

Local PDFs and source are review material. Scientific review, publication authority and a complete release certificate are not supplied by a successful build.

Evidence: `reports/science/paper-accounts.md`; `reports/needed/README.md`; `docs/public/pollenomics/science/papers.md`